ECE 102 Homework 3

Edi Zhang

May 3, 2025

1. **Linear Systems**

   Determine whether each of the following systems is linear:

   (a) $y(t) = \cos(t)x(t)$

   Solution:

$$
\begin{aligned}
S(ax(t) + b\tilde{x}(t)) &= \cos(t)(ax(t) + b\tilde{x}(t)) && (1)\\
&= a\cos(t)x(t) + b\cos(t)\tilde{x}(t) && (2)\\
&= aS(x(t)) + bS(\tilde{x}(t)) && (3)
\end{aligned}
$$

   Thus, the system is linear

   (b) $y(t) = \frac{d}{dt}(\frac{1}{2}x(t)^2)$

   Solution:

$$
\begin{aligned}
S(ax(t) + b\tilde{x}(t)) &= \frac{d}{dt}\left(\frac{1}{2}(ax(t) + b\tilde{x}(t))^2\right) && (4)\\
&= \frac{d}{dt}\left(\frac{1}{2}((ax(t))^2 + (b\tilde{x}(t))^2 + 2abx(t)\tilde{x}(t))\right) && (5)\\
&= \frac{d}{dt}\left(\frac{1}{2}(ax(t))^2\right) + \frac{d}{dt}\left(\frac{1}{2}(b\tilde{x}(t))^2\right) + \frac{d}{dt}\left(2abx(t)\tilde{x}(t)\right) && (6)\\
&\neq aS(x(t)) + bS(\tilde{x}(t)) && (7)
\end{aligned}
$$

   The system is not linear

   (c) $y(t) = e^{x(t)}$

   Solution:

$$
\begin{aligned}
S(ax(t) + b\tilde{x}(t)) &= e^{ax(t)+b\tilde{x}(t)} && (8)\\
&= e^{ax(t)} \cdot e^{b\tilde{x}(t)} && (9)\\
&\neq ae^{x(t)} + be^{\tilde{x}(t)} && (10)
\end{aligned}
$$

The system is not linear

(d) $y(t) = x(t) + 2u(t+1)$

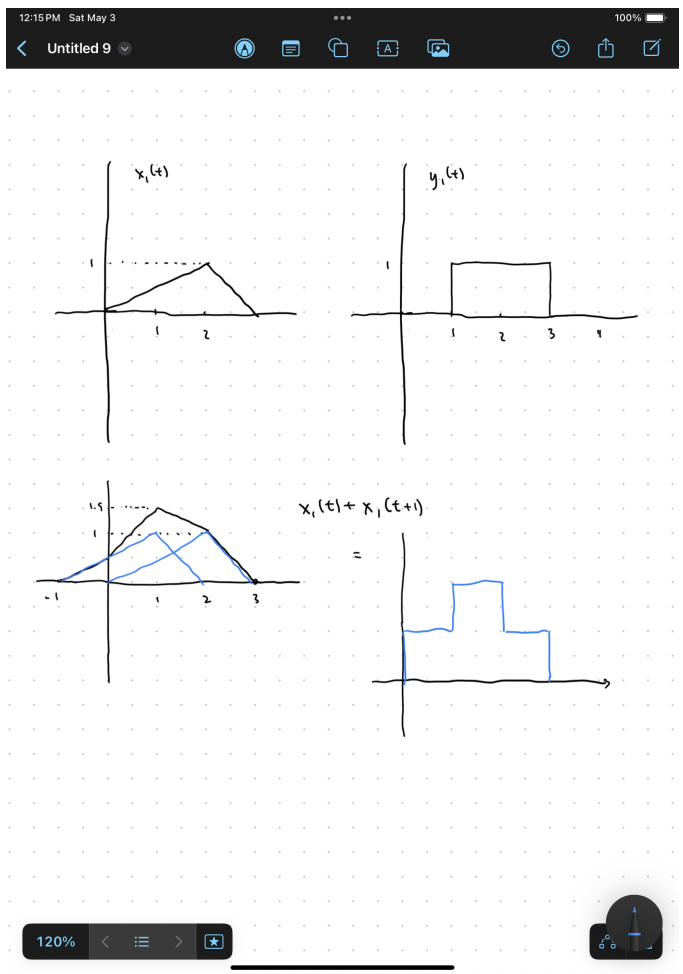
Solution:

$$S(ax(t) + b\tilde{x}(t)) = ax(t) + b\tilde{x}(t) + 2u(t+1) \tag{11}$$

The system is not linear. Additionally, $S(0) \implies$ the output should be 0 for a linear system. However, $y(t)$ contains a function $2u(t+1)$ independent of $x(t)$. Therefore the system would violate homogeneity

2. **LTI Systems**

(a) Consider an LTI (linear time-invariant) system whose response to $x_1(t)$ is $y_1(t)$, where $x_1(t)$ and $y_1(t)$ are illustrated as follows:



(b) Assume we have a linear system with the following input-output pairs:

- the output is $y_1(t) = \cos(t)u(t)$ when the input is $x_1(t) = u(t)$
- the output is $y_2(t) = \cos(t)(u(t+1) - u(t))$ when the input is $x_2(t) = \text{rect}(t + \frac{1}{2})$

Is the system time-invariant?

Solution:

Notice that $u(t+1) - u(t) = \text{rect}(t + \frac{1}{2}$. Assume $y(t) = \cos(t)x(t)$

$$H(ax(t) + b\tilde{x}(t) = \cos(t)(ax(t) + b\tilde{x}(t)) \tag{12}$$
$$= \cos(t)ax(t) + \cos(t)b\tilde{x}(t) \tag{13}$$
$$= aH(x(t)) + bH(\tilde{x}(t)) \tag{14}$$

We are given that $S(u(t)) = \cos(t)u(t)$ and $S(\text{rect}(t + \frac{1}{2}) = \cos(t)(u(t+1) - u(t))$.

Shifting the input,

$$y(t - t_0) = \cos(t)x(t - t_0) \tag{15}$$

The system is time-invariant.

3. **Convolution:**

(a) For each pair of the signals given below, compute their convolution using the flip-and-drag technique

i. $f(t) = \delta(t+1) + 2\delta(t-2), \quad g(t) = e^{-t}u(t)$

Solution:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \tag{16}$$
$$= \int_{-\infty}^{\infty} (\delta(t+1) + 2\delta(t-2))e^{-(t-\tau)}u(t-\tau)d\tau \tag{17}$$
$$= \int_{-\infty}^{\infty} \delta(t+1)e^{-(t-\tau)}u(t-\tau)d\tau + \int_{-\infty}^{\infty} 2\delta(t-2)e^{-(t-\tau)}u(t-\tau)d\tau \tag{18}$$
$$= e^{-(t+1)}u(t+1) + 2e^{-(t-2)}u(t-2) \tag{19}$$

ii. $f(t) = 2\text{rect}(t - \frac{3}{2}), \quad g(t) = 2r(t-1)\text{rect}(t - \frac{3}{2})$

Solution:

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \tag{20}$$

$f(\tau) \neq 0 \iff \tau \in [1,2]$ and $g(t-\tau) \neq 0 \iff t - \tau \in [1,2] \iff \tau \in [t-2, t-1]$.

Thus $[1,2] \cap [t-2, t-1] \neq \emptyset \iff \max(1, t-2) < \min(2, t-1) \iff t \in (2,3)$

For $t \in (2,3)$:

$f(\tau) \neq 0 \implies f(\tau) = 2$

$g(t-\tau) \neq 0 \implies g(t) = 2(t-\tau-1)$ Then

$$(f*g)(t) = \int_{\max(1,t-2)}^{\min(2,t-1)} f(\tau)g(\tau)d\tau \quad \text{for} \quad t \in (2,3) \tag{21}$$

$$= 4\int_{\max(1,t-2)}^{\min(2,t-1)} (t-\tau-1)d\tau \tag{22}$$

Notice that we can just integrate from $t-2$ to $t-1$, because non-overlapping regions get zeroed out. Thus

$$(f*g)(t) = 4\int_{t-2}^{t-1} (t-1-\tau)d\tau \tag{23}$$

$$= 4\left[(t-1)\tau - \frac{1}{2}\tau^2\right]_{t-2}^{t-1} \tag{24}$$

$$= 4\left[(t-1)^2 - \frac{1}{2}(t-1)^2 - (t-1)(t-2) + \frac{1}{2}(t-2)^2\right] \tag{25}$$

$$= 4\left[\frac{1}{2}(t-1)^2 - (t-1)(t-2) + \frac{1}{2}(t-2)^2\right] \tag{26}$$

Thus the convolution is

$$(f*g)(t) = \begin{cases} 2\left[(t-1)^2 - 2(t-1)(t-2) + (t-2)^2\right] & t \in (2,3) \\ 0 & \text{otherwise} \end{cases} \tag{27}$$

(b) For each of the following, find a function $h(t)$ such that $y(t) = x(t) * h(t)$

i. $y(t) = \int_{t-T}^{t} x(\tau)d\tau$

Solution:

We want

$$\int_{t-T}^{t} x(\tau)d\tau = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{28}$$

Therefore $h(t-\tau) = 1$ when $\tau \in [t-T, t]$ and 0 otherwise.

We have that $\tau \in [t-T, t]$

$$\iff t-T \leq \tau \leq t \tag{29}$$

$$-T \leq \tau - t \leq 0 \tag{30}$$

$$0 \leq t-\tau \leq T \tag{31}$$

Let $s = t - \tau$. Thus, we want $h(s) = 1$ for $s \in [0, T]$ and 0 otherwise.

$$h(s) = \begin{cases} 1, & s \in [0, T] \\ 0, & \text{otherwise} \end{cases} \tag{32}$$

Thus $h(t) = u(t) - u(t - T)$.

ii. $y(t) = \sum_{n=-\infty}^{\infty} x(t - nT_s)$

Solution:

We want

$$\sum_{n=-\infty}^{\infty} x(t - nT_s) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \tag{33}$$

Notice we are adding discrete values of some periodic function $x(t)$. Therefore we want $h(t)$ to zero out all the values of $x(t)$ that aren't $t - nT_s$ for $n \in \mathbb{Z}$.

We want to "pick out" $x(t - nT_s)$ for each $n \in \mathbb{Z}$. Thus we should use the Dirac Delta function and the sifting property. Recall that

$$\int_{-\infty}^{\infty} f(\tau)\delta(\tau - a)d\tau = f(a) \tag{34}$$

Therefore,

$$\int_{-\infty}^{\infty} x(\tau)\delta(\tau - (t - nT_s))d\tau = x(t - nT_s) \tag{35}$$

Thus,

$$\int_{-\infty}^{\infty} \left(x(\tau) \sum_{n=-\infty}^{\infty} \delta(\tau - (t - nT_s)) \right) d\tau = \sum_{n=-\infty}^{\infty} x(t - nT_s) \tag{36}$$

$$\int_{-\infty}^{\infty} \left(x(\tau) \sum_{n=-\infty}^{\infty} \delta(nT_s - (t - \tau) \right) d\tau = \sum_{n=-\infty}^{\infty} x(t - nT_s) \tag{37}$$

Hence,

$$h(t - \tau) = \sum_{n=-\infty}^{\infty} \delta(nT_s - (t - \tau)) \tag{38}$$

$$\implies h(t) = \sum_{n=-\infty}^{\infty} \delta(nT_s - t) \tag{39}$$

(c) Simplify the following expressions:

i. $[\delta(t - 3) + \delta(t + 2)] * [e^{3t}u(-t) + \delta(t + 2) + 2]$

Solution:

Recall that

$$(f+k)*(g+h) = ((f+k)*g) + ((f+k)*h) \tag{40}$$
$$= (f*g) + (k*g) + (f*h) + (k*h) \tag{41}$$

Let $f = \delta(t-3), \quad k = \delta(t+2), \quad g = e^{3t}u(-t), \quad h = \delta(t+2)+2$
Then,

$$(f*g) = e^{3t-9}u(3-t) \tag{42}$$

$$(k*g) = e^{3t+6}u(-t-2) \tag{43}$$

$$(f*h) = \delta(t-3)*[\delta(t+2)+2] \tag{44}$$
$$= \delta(t-3)*\delta(t+2) + \delta(t-3)*2 \tag{45}$$
$$= \int_{-\infty}^{\infty} \delta(t-3)(\delta(t-\tau+2)d\tau + 2\int_{-\infty}^{\infty} \delta(t-3)d\tau \tag{46}$$
$$= \delta(t-1)+2 \tag{47}$$

$$(k*h) = \delta(t+2)*[\delta(t+2)+2] \tag{48}$$
$$= \delta(t+4)+2 \tag{49}$$

Thus the final expression becomes

$$e^{3t-9}u(3-t) + e^{3t+6}u(-t-2) + \delta(t-1) + \delta(t+4) + 4 \tag{50}$$

ii. $\frac{d}{dt}[(u(t)-u(t+1))*u(t-2)]$ Hint: Show first that $u(t)*u(t) = r(t)$ where $r(t)$ is the ramp function:
Solution:

$$u(t)*u(t) = \int_{-\infty}^{\infty} u(\tau)u(t-\tau)d\tau \tag{51}$$
$$= \int_{0}^{t} u(\tau)u(t-\tau)d\tau \tag{52}$$
$$= \int_{0}^{t} 1\cdot 1 d\tau \quad \text{for } t \geq 0 \tag{53}$$
$$= t \quad \text{for } t \geq 0 \tag{54}$$

If $t < 0$, then $u(t-\tau) = 0$ unless τ is very negative, which would then make $u(\tau) = 0$. Thus $u(t) * u(t) = 0$ for $t < 0$. Thus

$$u(t) * u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases} = r(t) \tag{55}$$

Next,

$$(u(t) - u(t+1)) * u(t-2) = u(t) * u(t-2) - u(t+1) * u(t-2) \tag{56}$$
$$= r(t-2) - r(t-1) \tag{57}$$

Differentiating, we get

$$\frac{d}{dt}[r(t-2) - r(t-1)] = u(t-2) - u(t-1) \tag{58}$$

(d) Explain whether each of the following statements is true or false.

i. If $x(t)$ and $h(t)$ are both odd functions, and $y(t) = x(t) * h(t)$, then $y(t)$ is an even function.

Solution:

We want to check if $y(-t) = y(t)$

$$y(-t) = \int_{-\infty}^{\infty} x(\tau)h(\tau - t)d\tau \tag{59}$$
$$= \int_{-\infty}^{\infty} x(\tau) \cdot -h(t-\tau)d\tau \quad h(t) \text{ odd} \tag{60}$$
$$= -\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{61}$$
$$= -x(t) * h(t) \tag{62}$$
$$= -y(t) \tag{63}$$

The statement is **not true**.

ii. If $y(t) = x(t) * h(t)$, then $y(2t) = h(2t) * x(2t)$

Solution:

We are given

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{64}$$

LHS:

$$y(2t) = \int_{-\infty}^{\infty} x(\tau)h(2t-\tau)d\tau \tag{65}$$

Let $\tau = 2\lambda$. Then $d\tau = 2d\lambda$. Substituting, we get

$$y(2t) = \int_{-\infty}^{\infty} x(2\lambda)h(2t - 2\lambda)2d\lambda \tag{66}$$
$$= 2\int_{-\infty}^{\infty} x(2\lambda)h(2(t - \lambda))d\lambda \tag{67}$$
$$= 2(x(2t) * h(2t)) \tag{68}$$

Thus the statement is **not true**

4. **Impulse Response and LTI Systems** Consider the following three LTI systems:

- The first system S_1 is given by its input-output relationship: $y(t) = \int_{-\infty}^{t} x(\tau - t_0)d\tau$;
- The second system S_2 is given by its impulse response $h_2(t) = u(t - 2)$;
- The third system S_3 is given by its impulse response: $h_3(t) = u(t + 3)$

(a) Compute the impulse responses $h_1(t)$ of system S_1.
Solution:

$$h_1(t) = \int_{-\infty}^{t} \delta(\tau - t_0)d\tau \tag{69}$$

Thus

$$h_1(t) = \begin{cases} 1, & t \geq t_0 \\ 0, & t < t_0 \end{cases} \tag{70}$$
$$= u(t - t_0) \tag{71}$$

(b) The three systems are interconnected. Determine the impulse response $h_{eq}(t)$ of the equivalent system.
Solution:

$$S_1(x(t)) - S_3(S_2(x(t))) \tag{72}$$

Therefore

$$h_{eq} = S_1(\delta(t)) - S_3(S_2(\delta(t))) \tag{73}$$
$$= u(t - t_0) - S_3(u(t - 2)) \tag{74}$$
$$= u(t - t_0) - u(t + 3) * u(t - 2) \tag{75}$$
$$= u(t - t_0) - (t + 1)u(t + 1) \tag{76}$$

(c) Determine the response of the overall system to the input $x(t) = \delta(t) + \delta(t-3)$.

Solution:

$$h_{eq}(x(t)) = h_{eq}(\delta(t) + \delta(t-3)) \tag{77}$$

$$= h_{eq}(\delta(t)) + h_{eq}(\delta(t-3)) \tag{78}$$

$$= u(t-t_0) - (t+1)u(t+1) + u(t-t_0-3) - (t-2)u(t-2) \tag{79}$$